Natural Language Processing

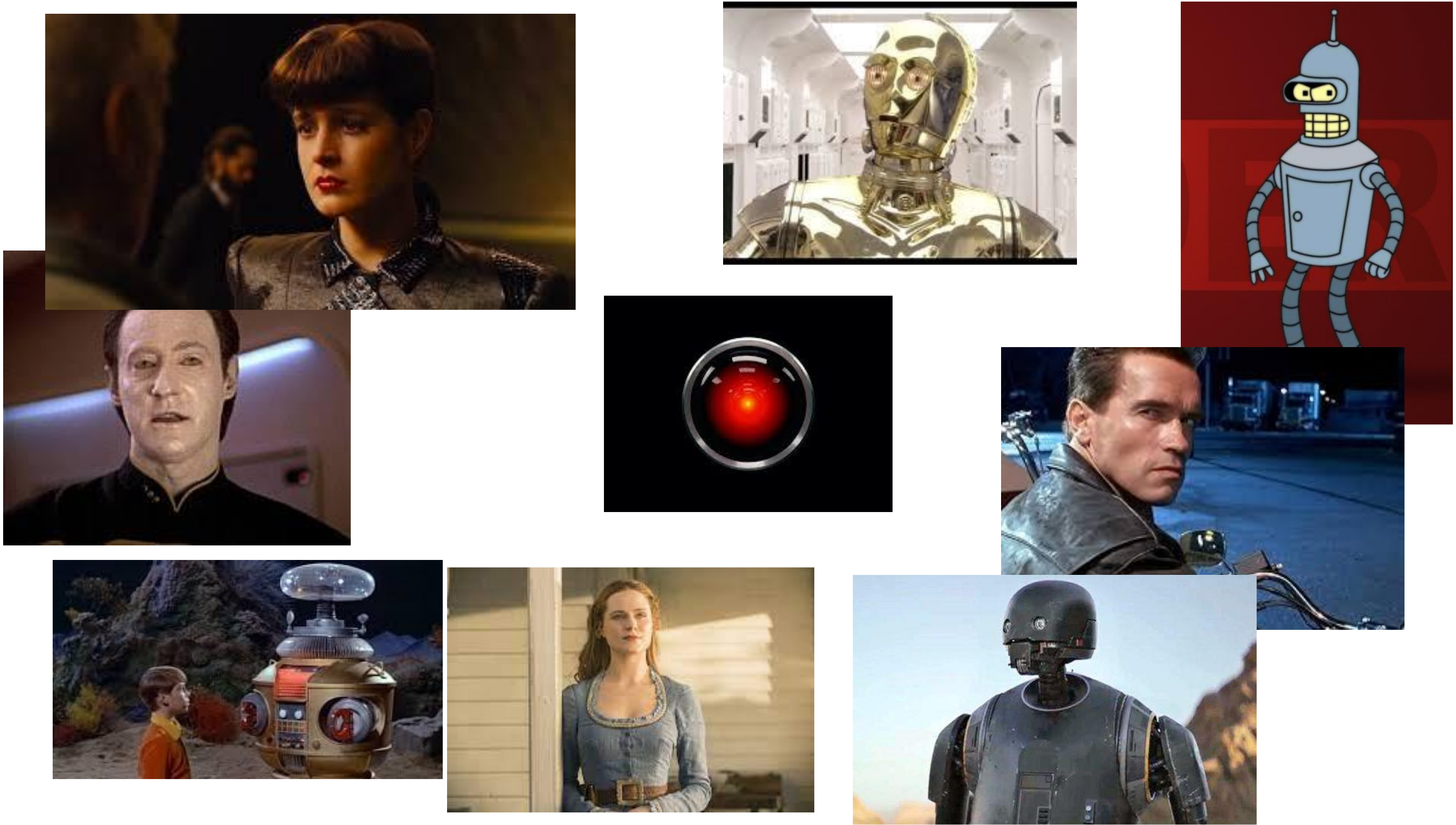
Jim Martin -- CSCI 5832

Lecture 1

Boulder NLP

Today

- Intro to NLP
- Logistics
- Some linguistics and a start on the lexicon



alexa

Google Assistant

Siri

Claude

LLaMA

ChatGPT

MISTRAL AI_

Gemini

Applications

Cointelegraph

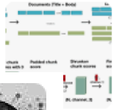
How to detect fake news with natural language processing

Discover how natural language processing (NLP) can be used to
news. Learn about the key techniques, examples and...

4 weeks ago

Nature

Identifying COVID-19 cases and extracting patient reported symptoms from Reddit using natural language processing



Tripwire

The Use of Natural Language Processing for Identifying and

complex, and widespread, posing a
nsitive...

Regenstrief Institute

Natural language processing extracs social risk factors from EHRs

New system from Regenstrief Institute and Indiana University overcomes challenges of generalizability and portability.

4 days ago

Nat
oph

SAN DIEGO — Natural language processing could help streamline documentation and improve patient care, according to a presentation at DOS...

May 8, 2023

EnterpriseAI

Predicting the Financial Market with Large Language Models

Large Language Models (LLMs) are made of artificial neural networks associated with millions or billions of parameters and trained on...

Jun 29, 2023

Government Technology

Illinois Adopts Natural Language Processing Tech for Child Welfare

The Illinois Department of Children and Family Services has adopted Augintel's Natural Language Processing software to help child welfare...

2 weeks ago

Unlocking the Power of Natural Language Processing in Identifying Social Risk Factors

A new study conducted by the Regenstrief Institute and Indiana University Richard M. Fairbanks School of Public Health has demonstrated the...

Natural Language Processing

NLP is the study of what goes into getting computers to perform *useful* and *interesting* tasks involving human language.

Computational linguistics is the use of computer models to provide *insight* into the science of language.

Why Now?

- An enormous an gital form
  - Newspapers, we filings, scientific articles, user manuals, fi rums, social media, etc.
- Advances in spee ational agents a key form of human-n
  - Especially in con al
- Much of human- diated by computers and is therefore sis
  - Reddit, faceboo

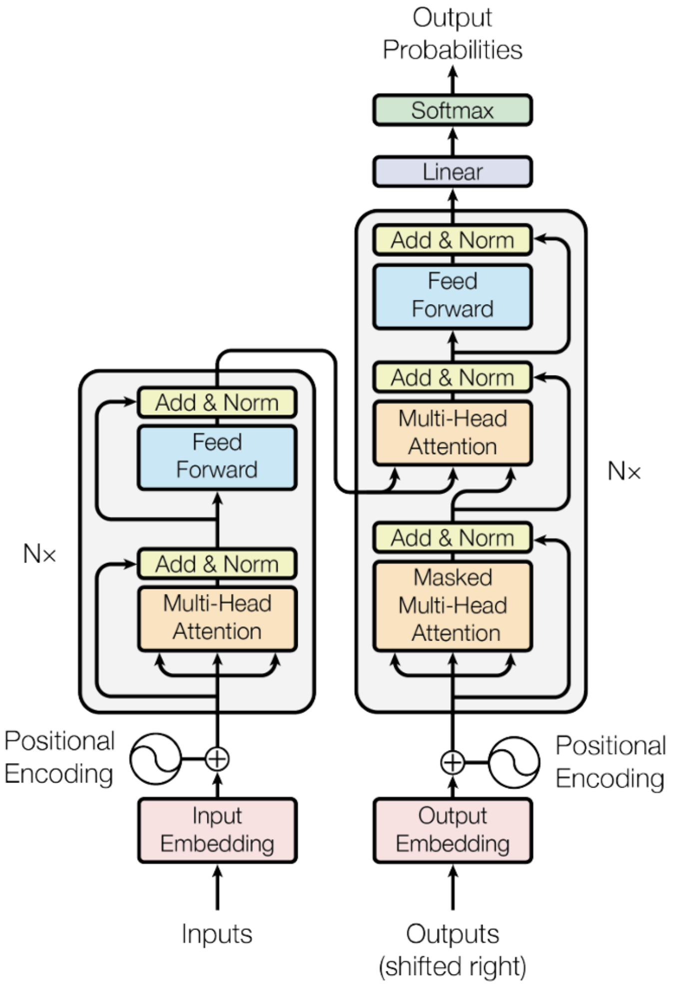


Figure 1: The Transformer - model architecture.

Why is NLP Important Now?

Uniform generative methods increasingly allow general-purpose models to create useful outputs for many specific applications.

- Machine translation
- Dialog
- Question answering
- Document generation
  - Scripts, reports, medical summaries, …

Topics

In this class we will intermingle discussions of

- Linguistics
  - Words, syntax, semantics, discourse, dialog
- Model architectures and assumptions
  - Mostly machine learning using NNs and embeddings
- Frontier models, applications and evaluations

Pillars of Natural Language Processing

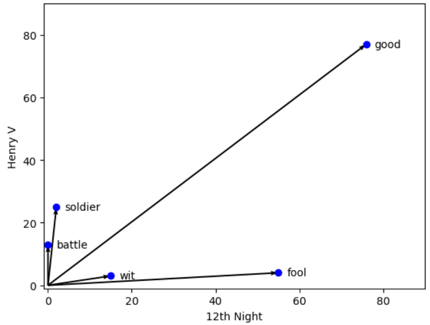
Probabilistic Language Models

$$P(w_{1:n})$$

Learning with Neural Networks

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \mathcal{L}(w)$$

Embeddings



Administrative Stuff

- Course home
  - canvas.colorado.edu
- Sections
  - There are 2 sections of the CSCI course and 1 Ling section
    - They are all the same. Same requirements, same grading, etc.
  - Currently there is no waitlist so everyone should be ok.

Course Staff

- James Martin – Instructor
- Enora Rice – TA
- Maria Valentini – TA

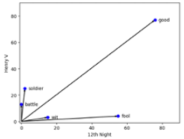
Preparation

- Ability to program in Python
- Basic algorithm and data structure analysis
- Exposure to basic concepts in
  - Discrete probability
  - Calculus
  - Linear algebra
- Familiarity with linguistics, psychology, and philosophy
- Exposure to formal logic
- Ability to write in English

Probabilistic Language Models $P(w_{1:n})$

Learning with Neural Networks $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \mathcal{L}(w)$

Embeddings



Machine Learning

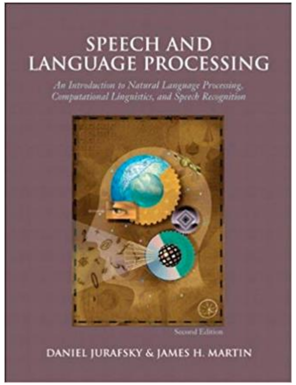
Supervised machine learning approaches, and neural networks in particular, are the dominant way to to build NLP systems

- This typically involves the use of large amounts of unlabeled data for training, and smaller amounts of curated data for finetuning and evaluation

- This is not an ML/Deep Learning class. If you’ve had an ML class that will be an advantage. But it’s not required.

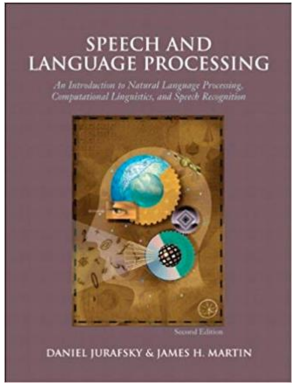
Requirements

- Readings
  - *Speech and Language Processing* by Jurafsky and Martin. Prentice-Hall <u>(Do NOT purchase the 2nd edition)</u>
  - Draft chapters that are in prep for a new 3rd edition are online.
  - Available here: https://www.stanford.edu/~jurafsky/slp3
  - We just dropped a new release



Requirements

- Readings
  - *Speech and Language Processing* by Jurafsky and Martin. Prentice-Hall (Purchase NOT required)
  - Draft chapters that are in prep for a new 3rd edition are online.
  - https://www.stanford.edu/~jurafsky/slp3
- 3 or 4 assignments
  - Both programming and written
- Shared task (SemEval)
- 3 quizzes
- In-class clicker responses



Shared Task

A "shared task" is a public NLP/ML task where a task definition, data, and evaluation metrics are made available. Groups are free to join the task and contribute towards understanding the problem.

Here, we'll provide options selected from the upcoming SemEval 2027 tasks and groups of 2-3 will choose a task and submit for the assignment.

SemEval-2026 Tasks | SemEv

semeval.github.io/SemEval2026/tasks

Bookmarks | Speech and Langu... | James Martin - Ho... | Meeting Schedule... | Computer Science... | Transformers | Logistic Regression | Logistic regressio... | What Is a Dynamic...

View on GitHub

SemEval-2026

The 20th International Workshop on Semantic Evaluation

We are pleased to announce the following tasks for SemEval-2026!

TASKS

Websites and contact information for individual tasks will be given below soon.

Sentiment and Emotion

- **Task 1: Humor Generation** ([contact organizers], [join task mailing list])
  Santiago Castro, Ignacio Sastre, J. A. Meaney, Luis Chiruzzo, Naihao Deng, Rada Mihalcea, Salar Rahili, Santiago Góngora, Aiala Rosá, Guillermo Moncecchi and Juan José Prada
- **Task 2: Predicting Variation in Emotional Valence and Arousal over Time from Ecological Essays** ([contact organizers], [join task mailing list])
  Nikita Soni, H. Andrew Schwartz, Ryan L. Boyd, Tony Bui, Syeda Mahwish, August Håkan Nilsson, Adithya V Ganesan, Lyle Ungar, Niranjan Balasubramanian and Saif M. Mohammad
- **Task 3: SemEval-2026 Task Proposal: Dimensional Aspect-Based Sentiment Analysis (DimABSA)** ([contact organizers], [join task mailing list])
  Liang-Chih Yu, Lung-Hao Lee, Jin Wang, Jan Philip Wahle, Terry Ruas, Kai-Wei Chang, Shamsuddeen Hassan Muhammad and Saif M. Mohammad

Narrative

- **Task 4: Narrative Story Similarity and Narrative Representation Learning** ([contact organizers], [join task mailing list])
  Hans Ole Hatzel, Ekaterina Artemova, Haimo Stiemer, Natalia Fedorova, Evelyn Gius and Chris Biemann
- **Task 5: Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding** ([contact organizers], [join task mailing list])
  Janosch Gehring and Michael Roth

QA and Conversations

- **Task 6: CLARITY - Unmasking Political Question Evasions** ([contact organizers], [join task mailing list])
  Konstantinos Thomas, George Filandrianos, Maria Lymperaiou, Chrysoula Zerva and

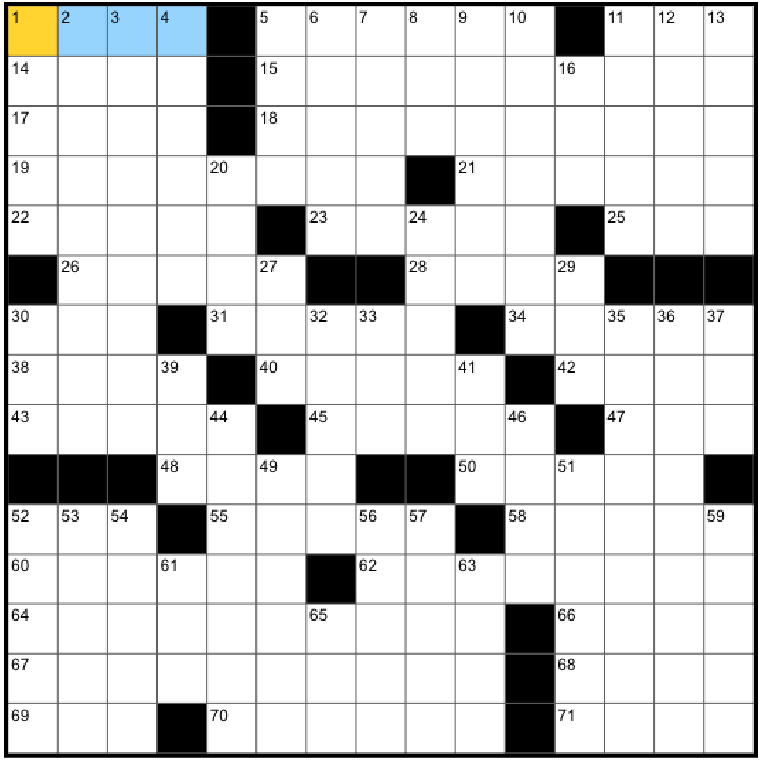
Grading

- Quizzes – 45%
- Shared Task – 20%
- Assignments -- 15%
- Clicker responses – 15%
- Engagement – 5%

Questions?

Words

1A Cleric's closetful



ACROSS

1 Cleric's closetful

5 One working on a tablet, say

11 Common creature in rebus puzzles

14 It's on the hook

15 Google's streaming device

17 Juice veggie

18 Paradise

19 Attempts to be a team player?

21 Accessory for dinner and a show?

22 Bland

23 Antique tint

25 Night class, perhaps, in brief

26 Parts of a spine

28 Kind of adapter

30 "Big" thing overseas

31 Smooths, in a way

34 Hive-minded?

38 Mars' counterpart

40 Proverbial assessment for

DOWN

1 "Grey's Anatomy" airer

2 Half of the only mother/daughter duo to be nominated for acting Oscars for the same film

3 Targets of some waxing

4 Inures

5 Bounce off the walls

6 Country with more immigrants than any other, informally

7 Fruit container

8 Kind of lane

9 Poison control remedy

10 Home of more than 16,000 slot machines

11 Fitness class inspired by ballet

12 "Cómo ____?"

13 Kind of alcohol that is a fermented biofuel

16 Litter maker

20 Bladders, e.g.

24 No longer done

Words/Tokens

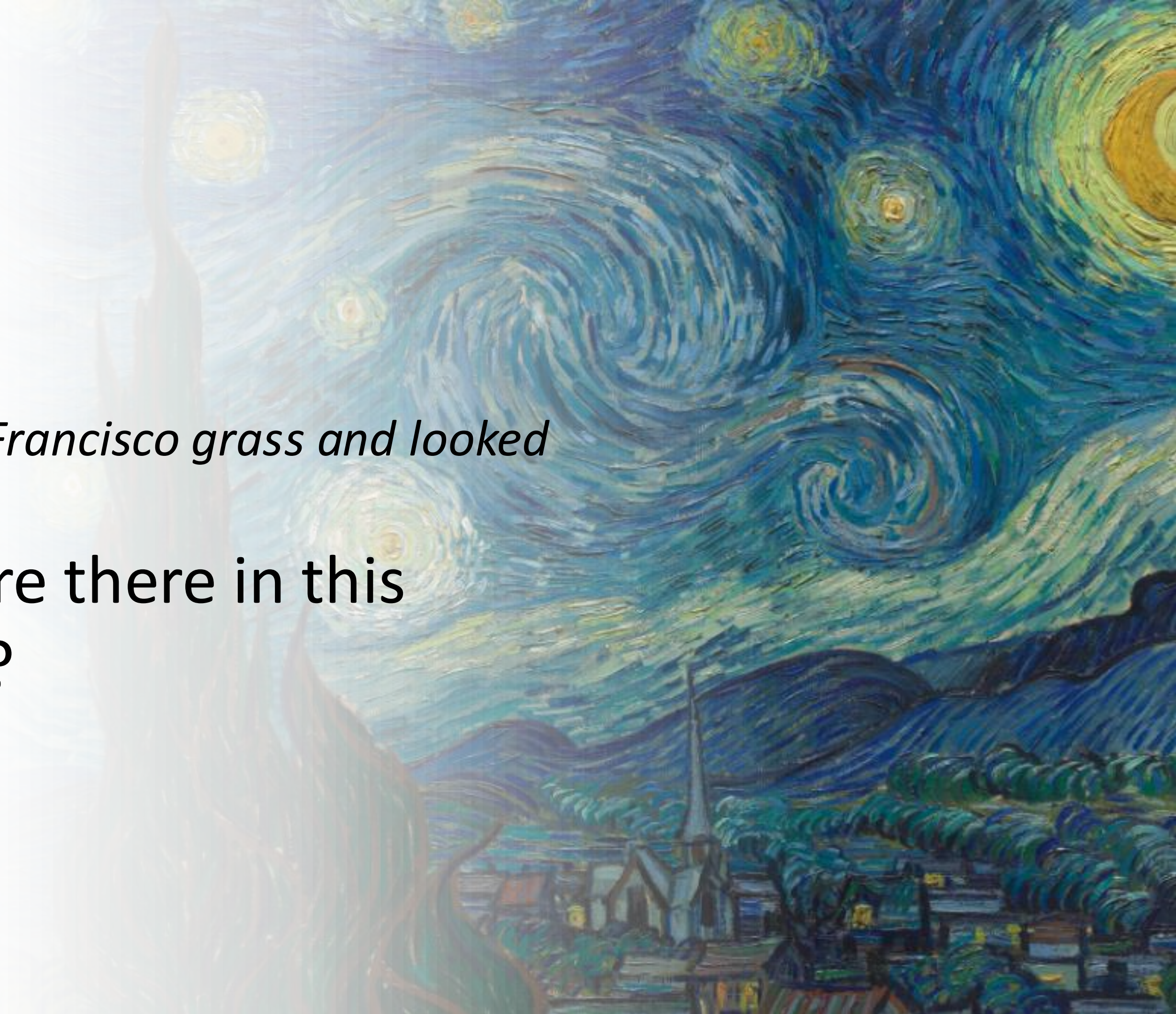
- In modern NLP systems, words/tokens are the fundamental bridge from linguistic form to underlying conceptual representations.
- Managing words/tokens and their representations will be the first problem that we’ll tackle.

Words

They lay back on the San Francisco grass and looked at the stars and didn't…

How many words are there in this sentence fragment?

- 15
- 14
- 13
- 16



Words

They lay back on the San Francisco grass and looked at the stars and didn’t

- *Type*: an entry in a vocabulary (or a lexicon)
- *Mention or token*: an instance of a type in some text

Words

They lay back on the San Francisco grass and looked at the stars and didn't…

- *Type*: an entry in a vocabulary
- Would you expect *looked* and *stars* to be listed in a vocabulary?

Words

They lay back on the San Francisco grass and looked at the stars and didn't…

- *Type*: an entry in a vocabulary
- Would you expect *looked* and *stars* to be in an English dictionary/vocabulary?
- No. You'd expect entries for *look and star*, you get *looked, looking, looks, starring, stars, starred, and starry* for free.

Of course, nothing comes for "free". In this case, you get all these forms from morphology.

Morphology

- *Morphology* is the study of the ways that words are built up from smaller units called morphemes
  - *Morphemes are the minimal meaning-bearing* units in a language
- We can divide morphemes into two classes
  - Stems: The core meaning-bearing units
  - Affixes: Bits and pieces that adhere to stems to change their meanings and grammatical function
  - So with “star_s”.   “star” is a stem and ”s” is an affix that signals a plural. Both are morphemes.

Inflectional Morphology

Inflectional morphology concerns the combination of stems and affixes where the resulting word....

- Has *the <u>same word class</u>* as the original
- And serves a grammatical/semantic purpose that is
  - <u>*Different*</u> from the original
  - But is nevertheless <u>*transparently* related</u> to the original
    - “walk” + “ing” = “walking”
    - “cat” + “s” = “cats”

Word Classes

Traditional parts of speech

- Noun, verb, adjective, preposition, adverb, article, interjection, pronoun, conjunction, etc.
  - There are various names for this notion
    - Part of speech, lexical category, word class, morphological class, lexical tag...

Word Classes: Parts of Speech

Three sources of evidence for part of speech categories

1. Semantics
2. Morphological evidence
3. Distributional evidence

What’s a noun?

Word Classes: Parts of Speech

- Three sources of evidence
  1. ~~Semantics~~
  2. Morphological evidence
     - walk, walking, walked, walks (-ing, -ed, -s)
       - Probably a verb! Why? Because verb is the name we give to the category of words that undergo these changes!
  3. Distributional evidence

Word Classes: Parts of Speech

Three sources of evidence

1. ~~Semantics~~
2. Morphological evidence
   - walk, walking, walked, walks
     - Probably a verb
3. Distributional evidence
   - The <u>crash</u>, A crash, Two <u>crashes</u>, The big <u>crash</u>…
     - Probably a noun since nouns follow determiners and adjectives
     - Also followed by verbs.

For Next Time

- More about words, tokens and lexicons
- Read
  - SLP 3rd Ed. Chapters 1 and 2